Pranjal Arya

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Skills

- **Languages:** C, ARM Assembly, Bash scripting.
- **Protocols:** UART, I2C, SPI, SD Card, SDIO, USB 2.0.
- **Processor:** ARM Cortex - (M4, M33).
- **Miscellaneous:** Git, GDB, Saleae analyzer, SEGGER SystemView, FreeRTOS, TrustZone for Cortex-M.

Experience

Member of the platform software team responsible for drivers, BSP, and middleware development for PSoC6 and the upcoming family of Infineon PSoC devices.

Infineon Technologies | **Senior Software Engineer** | **April 2023 - Present**

- **Firmware development & bring-up**:
  - Developed bare-metal drivers and BSPs for various ARM Cortex-M4 and M33 based MCUs.
  - Performed Pre-silicon and post-silicon firmware bring-up.
  - Ensured smooth integration of delivered assets to middlewares, RTOS.

Infineon Technologies | **Software Engineer** | **July 2020 - March 2023**

- **Drivers**: Added features in UART, SPI, I2C, GPIO, SD Card, SDIO, and USB 2.0 HID drivers and middlewares to support ARM Cortex-M4 based chips. Analyzed critical issues reported by customers and internal teams and provided fixes.
- **BSP and CI**: Created scripts and makefiles to automate build and basic sanity testing. Integrated this framework in CI.
- **Middleware**: Upgraded cryptography hardware acceleration to support latest MbedTLS release on public Github (bit.ly/prnjl).

Cypress Semiconductor | **Intern** | **January 2020 - June 2020**

- **Internship project**: Unified peripheral drivers to support all PSoC6 devices and created a common peripheral driver library.

Education

B.Tech. (Electronics & Communication Engineering) - Nirma University, Ahmedabad — CPI: **8.24** | July 2016 - July 2020
HSC — **88.9%** | 2016
SSC — **89.7%** | 2014

Projects

- **Custom Real-Time Operating System development from scratch** — GitHub project link
  Developed a custom RTOS kernel and added support for Cortex-M architecture with the following features:
  - Co-operative scheduler, Round-Robin scheduler
  - Support to calculate the CPU Utilization
  - Heap memory manager to address the problem of memory fragmentation and to report memory usage statistics
  - Spinlock and semaphore support for synchronization
  - Mailbox for communication
- **Garbage collector library for C programs** — GitHub project link
  Designed and implemented a generic C library to detect memory leaks. It can parse the application's data structures and manipulate them to track the memory blocks malloc'd by the application.
- **Custom bootloader** — GitHub project link
  Developed a custom bootloader for STM32L476RG MCU with the following features:
  - Host application in PC interacts with a custom bootloader over UART.
  - Capable of maintaining version information and updating application firmware using an A/B swap mechanism.
  - Integrated a secure boot mechanism to verify the authenticity of the firmware image.
- **CPU Faults on Arm Cortex-M** — GitHub project link
  To understand ARM processor internals, generated various Arm Cortex-M4 system exceptions using software on TM4C123GXL Tiva C series MCU. Wrote a fault handler to dump the faulting stack information over debug prints.

Awards

- **Global Spot Award** — Infineon Technologies, January 2023
  Awarded for significant contribution to peripheral driver library development.
- **YIP Award (Infineon Idea Management Program)** — Infineon Technologies, February 2023
  Awarded to a team of 2 members for internal framework improvement resulting in 40% cost reduction.